

CHANNEL 1 SUBSYSTEM AND SWAPPER (CHNSYS,20) 3/15/73

ASC=IOT 51
 DSC=IOT 50
 EEM=IOT 4074
 ESM=IOT 55
 FBC=IOT 2015
 ISB=IOT 52
 LEM=IOT 74
 LSM=IOT 54
 RCK=IOT 32
 RNM=IOT 66
 RRI=IOT 37
 RRO=IOT 17

IOPTSU=106 /SYMBOLS COMMON IN CORES 0,4,15,17
 TTBL=156224
 FTBL=156232
 BITTB1=6400 4BITTB1=64
 BITTB2=6500
 BITTB3=6600
 BITTB4=6700
 FRELST=157000

I I I/ OFFSET DCH

/RE INITIALIZE THE EXECUTIVE

EEM
 LAC I DCORE
 SAR 2S
 IOR B1
 DCH (ADD JDA .+1)
 RNM
 CLC
 DAC ONTRACK
 LAC (XCT 27)
 DAC I DSWAPX
 LAC (210000) /START HERE FIRST TIME (30012)
 DIP NDONC2 1
 LIO (30000 NDONC2) /RE-INITIALIZE FRELST TABLE
 JSP DGO /READ AND WAIT FOR COMPLETION
 JMP I INIT

CH1, RRI /IS CHANNEL ABNORMAL

SPI

JMP I ABNORMAL

CH1+3, LIO CHNFLG /RESET FLAGS

LFI

JMP I CH1X /RETURN

CH1X, SWAP

CH1XE, -0

/START THE CHANNEL ROUTINE

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.GO,      LIO 15SET      /ANY CHANNEL 15 ACTIVITY?
          SNI I
          JMP 15GO
GO,        DAC CH1X
          LIO I CH1X
GO1,      IDX CH1X
GO1+1,    DZM CH1XE      /SET POSSIBILITY OF ERROR INSTRUCTION
          RRO 500
          JSP 1HLT
          IDX NUMRRO      /USED BY 1 SEC CLOCK IF FASTRAND HUNG
CHANX,    LIF
          DIO CHNFLG
          JMP I CH1BRK
NUMRRO,   Ø

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/JSP WAIT1 JSP WAIT2
/ADDR OF COMMANDS JUST ASSEMBLED
/ADDR OF PLACE TO PUT JUMP
/IN WAIT2 FLAG 6=0-START CHANNEL

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WAIT2,    SZF I 6        /DO WE WANT TO JUST START THE CHANNEL
          JMP WAIT16      /YES
WAIT1,    DAC WAIT11
          LIO I WAIT11
          IDX WAIT11
          LAC I WAIT11
          DAC WAIT12
          LAC BØ          /PLACE JUMP INSTRUCTION
          IAI
          DAC I WAIT12
          IDX WAIT11
WAIT14,   DIO WAIT12
          RRI              /HOW IS CHANNEL
          RIR 5S
          SPI
          JMP WAIT13
          JSP WAIT15      /REMOVE SEQUENCE BREAK
          LAC WAIT11
          DAC CH1X
          LIO WAIT12      /RESTART THE CHANNEL
          JMP GO1+1

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/NORMAL EXIT IF CHANNEL IS BUSY

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WAIT,     RRI
          RIR 5S
          SPI I
WAIT15,   ISB 100
WAIT13,   DAC CH1X
          JMP CHANX

WAIT16,   DAC CH1X
          LIO I CH1X
          IDX CH1X
          JMP GO1          /GO START THE CHANNEL

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WAIT11,   Ø
WAIT12,   Ø

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CHNFLG, 0

/CHANNEL 1 SWAPPER
 /DCONT=0←PREVIOUSLY NOT BUSY
 /DCONT>0←HELD BY FASTRAND
 /DCONT<0←HELD BY MTCU

SWAP:

SWAPE1, CLL"U"SCF
 LAC (177777) /PICK UP NEEDED MASK
 LIO DCONT /IS THERE ANYONE TO FREE
 SNI
 JMP SWAPE3 /NO-GO GET SOMEONE
 SPI
 JMP I SWAPE2 /HELD BY MTCU
 AND CONTR1 /FASTRAND WANTS TO FREE A GUY
 ADD THREE
 SAD (FTBL 146)
 LAC (FTBL)
 SAS END1
 IOR B1
 DAC CONTR1
 JMP I SWAPE7

DCONT, 0

SWAPE5, LAC (SWAP) /RESET THE CH1 SWAPPER
 DAC CH1X
 CLC
 DAC SCORE
 DAC CH1XE
 DAC OVERFLOW
 DZM DCONT

PAGE 4

```
SWAPE3,   LAC I CONTR2
          SPA
          JMP SWAPE4       /YES
          LIO CONTR1      /NO-DOES FASTRAND WANT CHANNEL
          SPI I
          JMP I CH1BRK     /NO-ONE WANTS ANYTHING
          DAC DCONT       /MARK AS HELD (KNOW AC STRICTLY POSITIVE)
          JSP GETUSER     /GET USER INTO CORE

FUSER,    Ø
          LIO I CONTR1    /CHECK MODE OF ENTRY
          SPI
          JMP I SWAPE6     /METHOD 1
FSTRLS,   LIO SWPFLG      /LOAD FLAGS
          LFI
          DZM APRFLG      /RESET APR FLAG
          DZM PATFLG      /RESET PATIENT IDENTIFIER FLAG
          LAC FPAR

SWAPE9,   JMP .
SWAPE6,   Ø

/MTCU WANTS DATA CHANNEL
SWAPE4,   DAC DCONT       /SET HELD(KNOW AC STRICTLY NEGATIVE)
          JSP GETUSER

TUSER,    Ø
          LAC I FTAPE1
          DAP SWAP1Ø
          LAC I SWAP1Ø     /DO EXTRA LEVEL OF INDIRECT
          DAP SWAP1Ø
          CLI
          RCL 6S
          LFI
          JSP WAIT4       /ALL TAPE ROUTINES WISH USER IN CORE IMMEDIATELY
          JMP I SWAP1Ø
SWAP1Ø,   DCH I .

GETUSER,  DAC GETUX
          LAC I GETUX     /GET USER NO.
CSUB,     SUB MSTAT
          DAP 1STAT
          LAC I 1STAT     /CHANGE IOP WANTS AND HUNG BITS
          XOR CSUB        /UPPER 6 BITS=42
          DIP I 1STAT
          ISB 17ØØ       /TELL SWAPPER
CDZMI,    DZM I HOTFLG
GETU2,    IDX GETUX
          JMP I GETUX

GETUX,    Ø
1STAT,    DCH .
```

PAGE 5

/CHANNEL 4

```
4CHAN,      LIO CONTR1
             SPI
             JMP I 4CHAN1
             RIL 1S
             SPI I           /IS FASTRAND CPU BUSY
             JMP I 4CHAN1
             RRI 1000        /YES-IS FASTRAND PHYSICALLY BUSY
             SPI
             JMP I 4CHAN1    /YES - WE'LL COME BACK LATER
4CHAN2,      JMP FTRACK
```

/START FASTRAND ROUTINE

```
FGO,        0           /ENTER WITH POINTER TO HALF TRACK TABLE
             DAP 4CHAN2   /RESET RETURN
             LAC FGO      /CALCULATE TRACK WANTED
             DAC ONTRACK  /RESET TRACK POINTER
             SUB (FRELST
             SAR 2S
             IOR C400
             LIA
             RRO 100      /MOVE THE BOOM
             JSP FHLT
             JMP I 4CHAN1
```

/CHANNEL 4 BOOM HANDLER

FTRACK, LAW RING /SET TO GO TO CHANNEL 15 TO FIND BEST GUY

DAP CH15B

JSP 4DP15

LSM

/DO WE WANT TO USE CHANNEL 1

LIO DCONT

SNI

JMP FTRK3

/YES

ESM

LAC HTR

FTRK2, XCT .

JMP METH0

/GO TO DATA CHANNEL BY METHOD 0

JMP FSTRLS

/GO TO CHANNEL RESTORE TO METHOD 0

/TABLE FOR CHANNEL 4 DISPATCH

FTRK5, JDA 4RD /MOVE BOOM FOR READ

JMP 4WT /MOVE BOOM FOR WRITE

JDA 4RW /MOVE BOOM FOR REWRITE

CNOP, NOP /NOP FOR FIXED HEADS ETC.

JMP 4HW /MOVE BOOM FOR HELD WRITES

/TABLE FOR CHANNEL 1 DISPATCH

FTRK6, JDA READ

JSP WRITE

JDA REWRITE

NOP

JSP 1HW

```

/CHANNEL 15 ROUTINES TO FIND BEST GUY
RING,    LAC B5          /SET TO LOWEST QUEUE
          DAC RNGQUE
          LAC CONTR1
          DAC FTRK1      /GET POINTER TO TOP GUY
RING1,    DAP RNGPTR     /RESET POINTER TO GUY WE'RE LOOKING AT
          LAW 177
          AND I RNGPTR   /GET GUY'S PROGRAM NUMBER
          ADD C200       /GET STAT TABLE POINTER FOR HIM
          DAP RING2
          LIO I RING2
          CLA
          RCL 6S
          SAS C40
          CLI
          RIR 6S
          LAI
          SUB RNGQUE     /COMPARE WITH BEST QUEUE SO FAR
          SMA
          JMP RING3      /NOT BETTER GUY
          DIO RNGQUE     /BETTER GUY : RESET BEST QUEUE
          LAC RNGPTR     /SAVE POINTER TO BEST GUY
          DAC RNGFST
RING3,    LAW 3          /INCREMENT TO LOOK AT NEXT GUY
          ADD RNGPTR
          SAD (FTBL+146)
          LAC (FTBL)
          SAS END1       /ARE WE DONE
          JMP RING1      /NO : LOOP

          CLA           /DECODE BEST GUY
          LIO I RNGFST
          RCL 6S
          ADD (FTRK5)
          DAP FTRK2
          ADD (FTRK6-FTRK5)
          DAP FTRK4
          RIR 6S
          DIO FUSER
          LAC I FTRK1
          DAC I RNGFST
          IDX RNGFST
          IDX FTRK1
          LAC I RNGFST   /DECODE SECOND WORD
          DAC FPAR
          AND C777
          CMA
          DAC FBLOCK
          LAW I 777
          AND FPAR
          RAR 9S
          ADD (FRELST)
          DAC HTR
          AND (-177)
          DAC THIRD

```


LAC I FTRK1
 DAC I RNGFST
 IDX RNGFST
 IDX FTRK1
 LAC I RNGFST
 DAP SWAPE9
 DCH (CMI SWPFLG)
 LAC I FTRK1
 DAC I RNGFST
 JMP CH15BE

SWPFLG,	Ø
RNGPTR,	DCH I .
RING2,	DCH .
RNGQUE,	Ø
RNGFST,	Ø

/VARIABLES FOR BOOM

FBLOCK, 0
 HTRACK, 0
 THIRD, 0
 FTRK1, 0
 ONTRACK, -0
 TRACKA, -0
 TRACKB, -0
 TRACKC, -0
 TRACKD, -0
 FPAR, 0

/-NO. OF BLOCKS WANTED

/THIRD OF DRUM WANTED

/POINTER TO RETURN ADDRESS

/HALF TRACK WE ARE ON

/HALF TRACK FOR WRITE EACH THIRD

/REWRITE LOCATION

/INITIALLY DO WE WANT TO USE CHANNEL 1

FTRK3, ESM

JSP METH1

/GO TO CHANNEL 1

LAC HTR

FTRK4,

XCT .

JMP FSTRLS

/GETTING TO CHANNEL 1

METH1, DAC SWAPE6

/SAVE RETURN

LAC B0

IOR I CONTR1

DAC I CONTR1

METH0,

LAC (550000

/SET WANT DATAT CHANNEL BIT

LSM

DIP CONTR1

LAC DCONT

/IS CHANNEL NOW BUSY

SZA I

ISB 100

ESM

LAW FTRACK

DAP 4CHAN2

/RESET RETURN

JMP I 4CHAN1

/CHANNEL 15 ROUTINES
/TO START D.C. AND GO TO CHANNEL 15

```

15GO,      DAC 15GOX
           ISB 1500      /GO TO CHANNEL 15 WHEN DEBREAK
           DZM 15TEL     /SET POINTER TO SAY STARTING
           DZM 15SET     /RESET SWITCH
           LAC I 15GOX   /GET PLACE TO START THE CHANNEL
           DAC .+2
           JSP GO
           30000 .      /.+2
           LIO 15TEL
           IDX 15TEL     /TELL 15 THAT 1 IS DONE
           IDX 15GOX     /RESET RETURN
           SNI I         /IS CHANNEL 15 DONE
           JMP I 15GOX   /YES-EXIT
           DAC CH1X
           JMP CHANX     /NO-GO WAIT FOR SEQUENCE BREAK

15GOX,      Ø
15TEL,      Ø
15SET,      Ø

```

```

/DROP FROM 1 TO 15
15DRP,      DAC 15GOX
           ISB 1500      /ACTIVATE CHANNEL 15
           DAC 15TEL     /MARK 1 AS DONE
           DZM 15SET
           JSP WAIT13
           JMP I 15GOX

```

/CHANNEL 15 SERVICE ROUTINE

```

CH15A,      JSP CH15B    /DAP'ED INTO BY HIGHER ROUTINES
CH15AE,      LAW CH15B
           DAP CH15A     /RESET STARTING ADDRESS
           CLA
           LSM
           SAS 15TEL
           ISB 100
           IDX 15TEL     /MARK 15 AS DONE
           ESM

CH15B,      JSP CH15C
CH15BE,      LAW CH15C
           DAP CH15B     /RESET STARTING ADDRESS
           LSM
           LAC CDZMI     /SET BUSY BIT (CDZMI UPPER 6 BITS =35)
           DIP CONTR1    /ASSUME POINTS TO CORE 15
           ESM
           ISB 400       /REACTIVATE CHANNEL 4

CH15C,      JMP I CH15BK

```

PAGE 11

/CHANNEL 4 ROUTINE TO MOVE BOOM AND GO TO 15
/USE FACT THAT TOP 6 BITS OF "FRELST"=15

```
4GO15,      Ø
            DAC 4GOX
            LAC (FRELST)  /REMOVE THE BUSY BIT SO WONT' DO ANYTHING
            DIP CONTR1
            ISB 1500
            LAC 4GO15      /GO MOVE THE BOOM
            JDA FGO
            JMP I 4GOX     /DON'T COME BACK TILL ACTIVATE BIT SET
4GOX,      Ø
```

/CHANNEL 4 ROUTINE TO DROP TO 15

```
4DP15,      DAP 4CHAN2
            ISB 1500
            LAC (FRELST)  /REMOVE THE BUSY BIT
            DIP CONTR1
            JMP I 4CHAN1
```

/FASTRAND WAITING LIST
 /WORD 0: BIT 0=1+METHOD 1 OF D.C. ACCESS
 /BIT 1-5 ARE FUNCTION
 /BITS 11-17 ARE PROGRAM NUMBER
 /WORD 1: 0-7 ARE HALF TRACK
 /BITS 11-17 ARE NO OF BLOCKS WANTED
 /OR BITS 0-1 ARE THIRD
 /BITS 2-10 ARE 9 BIT USER CODE
 /BITS 11-17 ARE NO. OF BLOCKS WANTED
 /WORD 3 ARE FLAGS AND RETURN ADDRESS
 /ONLYC CHANNEL 16 ADDS TO THIS LIST
 /CAN NOT OVERFLOW TABLE

TRACK, 0
 DAC TAPE3
 JSP RELEASE
 LAC END1 /GET END OF LIST VALUE
 DAC TAPE1
 LAC 16USER /GET USER NO.
 IOR I TAPE3 /BITS SET UP PROPERLY ALREADY FOR FUNCTION
 DAC I TAPE1
 IDX TAPE1
 LAC TRACK /GET DATA WORD
 DAC I TAPE1
 IDX TAPE1
 LIF"U"SCF /GET FLAGS
 RCR 6S
 DIP I TAPE1
 IDX TAPE3 /GET RETURN ADDRESS
 DAP I TAPE1
 IDX TAPE1
 SAD (FTBL 146
 LAC (FTBL
 LIA
 LSM
 LAC CONTR1 /ARE WE BUSY
 SAD END1
 ISB 400
 IOR B1 /SET CPU BUSY BIT (MUST BE AN OR)
 DAC CONTR1
 DIO END1
 ESM
 JMP I R0

16USER, 0

TAPE3, 0
 TAPE1, 0
 16STAT, DCH .
 CONTR1, FTBL
 END1, FTBL

/CHANNEL 17 ROUTINES

/CHANNEL 17 NORMAL RETURN (REAL CORE IN AC)

```
17SWAP,  LIO I (IOR 50) /SAVE UP THE USER
        DIO I AC
        LIO I (IOR 51)
        DIO I IO
        LIO I (IOR 52)
        DIO I FLAGS
        RCK
        DIO TIME
        IOR (600000)
        LSM
        DAC SCORE
        CLI"U"CMI"U"CLA
        SAS 17TEL
        JMP 17SWP2
        DIO 17TEL
        ISB 100
17SWP2,  ESM
        JMP I SWORG
```

/WAIT FOR CORE ROUTINE

```
WAIT4,  DAC CH1X
        LAC SCORE      /IS THE CORE AROUND
        SAS MZERO
        JMP I CH1X     /SOMETHING HAS ALREADY HAPPENED
        DZM 17TEL     /NO-TELL ME WHEN IT COMES
        JMP CHANX
```

```
SCORE,  -0
17TEL,  -0
```

```
RELEASE, DAC RELX
        LAC B0
        LSM
        IOR I 16STAT
        DAC I 16STAT
        ESM
        ISB 1700
        DZM I HOTFLG
        JMP I RELX
```

```
RELX,  0
```

```
TIME,  0
```

START XX-JMP